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DATA DESCRIPTOR

A high-resolution perspective-view road image dataset for pothole detection

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Potholes are a critical infrastructure distress that damages vehicles and creates safety risks, motivating timely detection for effective maintenance. Automated detection via computer vision is a scalable solution, but its progress is hindered by the scarcity of high-quality, large-scale public datasets. Here we present HRP4K, a high-resolution, perspective-view road image dataset for developing and evaluating pothole detectors. Images are captured by vehicle-mounted full-frame mirrorless cameras across 1,100 km of urban and rural roads in China. The dataset comprises 6,003 images, including 4,003 positive images containing at least one pothole and 2,000 negative images of pothole-absent road surfaces. In total, HRP4K provides 7,217 pothole instances annotated with bounding boxes and is released in both YOLO and COCO formats. A human-in-the-loop pipeline involving algorithmic preprocessing, privacy anonymization, and iterative model-assisted annotation ensured consistent, high-fidelity labels. The data exhibit realistic long-tailed distributions of small and ultra-small potholes in visually complex scenes, providing a challenging benchmark for object detection. We report baseline results using six modern detectors to facilitate standardized comparison and benchmarking in automated infrastructure monitoring.

Background & Summary

The structural integrity of road networks is a cornerstone of modern economies, directly impacting transportation efficiency, vehicle operating costs, and driver safety. Pothole formation, a severe manifestation of pavement distress, indicates underlying structural degradation and poses a direct threat to vehicular safety¹. This distress can damage vehicles, increase fuel consumption and operating expenses, and precipitate serious, sometimes fatal, crashes. Timely and accurate detection of such damage is therefore essential for effective prioritization and cost-efficient maintenance planning².

Traditional manual or instrumented pothole surveys (e.g., laser profilers) are labor-intensive, time-consuming, and costly³. In contrast, automated pothole detection methods, especially those utilizing digital camera-based systems, have gained significant research interest owing to their low cost, deployment flexibility, and ease of integration with various measurement platforms⁴. Computer vision algorithms, particularly deep convolutional neural networks (CNNs) and vision transformer networks (ViTs), have achieved remarkable success in object detection and segmentation, making them well-suited for identifying potholes in digital imagery. However, the performance of such data-driven models depends critically on the quality and quantity of training and validation data⁵. Progress remains constrained by the lack of large-scale, diverse, and well-annotated publicly available pothole datasets⁶.

Existing datasets illustrate an unresolved trade-off between data fidelity and scale, as summarized in Table 1. On the one hand, datasets like Pothole-600⁷ and UDTIRI⁸ prioritize high-quality annotations but are limited in scope, thereby restricting the environmental generalization of models trained on them. Other datasets, such as GAPS⁹, offer high-resolution (1920×1080) imagery for multiple distress types, including potholes, cracks, and surface wear. While the resolution is advantageous for capturing fine details, the dataset contains only 184 pothole images. This scarcity hampers deep learning models from learning robust feature representation, leading to overfitting and poor generalization across diverse pothole types and road conditions. Similarly, the work by Nienaber *et al.*^{10,11} provides high-resolution (3680×2760) perspective-view images; however, the use of a wide-angle lens substantially reduces the effective pavement area to approximately 1078×680 pixels.

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Dataset	Pothole images	Pothole instances	Total Images	Capture devices	Resolution	View Captured	Annotation-level	Annotated Category
Pothole-600 ⁷	600	618	600	ZED Stereo camera	400 × 400	Top-down	Pixel level	pothole
UDTIRI ⁸	1,000	1,462	1,000	Various cameras	Various (e.g., 1200 × 630)	Wide view	Bounding box & Pixel level	Pothole
GAPS V2 ⁹	184	—	2,468	High-resolution Camera	1920 × 1080	Top-down	Bounding box	4 types (Cracks, Potholes, etc.)
Nienaber <i>et al.</i> ^{10,11}	2,311	5,943	5,676	GoPro Hero3	3680 × 2760	Perspective-view	Bounding box	Pothole/normal
RDD2022 ¹³	3,674	6,544	47,420	Smartphone and cameras	Various (e.g., 600 × 600)	Wide view	Bounding box	4 types (Cracks, Potholes, etc.)
Kumar <i>et al.</i> ³¹	329	—	681	—	Various (e.g., 804 × 534)	Wide view	Class label	Pothole/normal
SVRDD ¹²	627	918	8,000	—	1024 × 1024	Street view	Bounding box	6 types (Cracks, Potholes, etc.)
HRP4K (Ours)	4,003	7,217	6,003	A7 IV and A9 III	3840 × 2160	Perspective-view	Bounding box	Pothole

Table 1. Comparison of public road damage datasets.

Furthermore, this dataset is heavily imbalanced, containing few positive pothole instances. The SVRDD¹² dataset, comprising 627 pothole instances across 8,000 images, presents a different challenge due to its street-view perspective. On the other hand, RDD2022¹³ focuses on scale, containing over 47,000 images from multiple countries to enhance model generalization. This scale, however, comes at the expense of data heterogeneity, with inconsistent resolutions (e.g., 600 × 600) and perspectives captured using various consumer-grade devices. This heterogeneity necessitates downsampling images to a uniform low resolution for batch processing, inevitably degrading the high-frequency textural and boundary details. Consequently, models trained on these datasets struggle to distinguish small, incipient potholes from shadows, water patches, or tar repairs, limiting their applicability for early-stage intervention.

In practice, road-surface distress detection primarily relies on imagery captured by in-vehicle cameras¹⁴, such as dashboard- and roof-mounted systems. This approach offers a cost-effective and rapid method for pothole detection. However, such images are typically captured from an oblique angle and at a considerable distance from the pavement surface. Consequently, small to medium-sized potholes are projected into a minimal image area. They often occupy only a handful of pixels, resulting in significant loss of detail. Nevertheless, timely detection and repair of such potholes are critical to preventing progressive deterioration. Early intervention mitigates escalating maintenance costs, and elevated crash risk. Despite their valuable contributions, existing pothole datasets often lack sufficient scale, diversity, or resolution, hindering the development of robust and accurate detection models.

To address this gap, we introduce the high-resolution, large-scale pothole dataset (HRP4K), a publicly available resource developed from real-world driving scenes. We employ a high-resolution, low-noise full-frame mirrorless camera imaging setup that preserves fine texture and edge cues, thereby mitigating the detail loss inherent to low-clarity imagery. Figure 1 illustrates two examples from the HRP4K dataset. Although the potholes appear small in the full image context, the high-resolution, low-noise imagery retains fine-grained details such as surface texture, crumbling edges, and local shading. This demonstrates the dataset’s capability to capture diverse pothole morphologies across varied real-world conditions with high fidelity. The main contributions are as follows:

- 1) We provide a new dataset comprising 6,003 high-resolution (4K) images, including 4,003 positive samples that capture a wide variety of pothole instances and 2,000 negative samples of pothole-absent road surfaces. The images are collected over approximately 1,100 kilometers of varied road types, spanning urban arteries and rural lanes on both asphalt and concrete surfaces. Data are acquired using two high-quality mirrorless cameras with standardized optics and capture settings to ensure technical consistency.
- 2) We propose a human-in-the-loop annotation pipeline that reduces manual effort while improving labelling accuracy. The pipeline begins with automated deduplication of redundant samples, followed by data anonymization using two object detection models. An iteratively trained detector then generates semi-automated pre-annotations to assist human annotators. Each stage incorporates human review and refinement to ensure high-quality final annotations.
- 3) We establish a pothole detection benchmark using six object detection algorithms and analyse their performance across diverse pavement types. Standardized annotations are provided in YOLO¹⁵ and COCO¹⁶ formats, enabling seamless integration with common deep learning pipelines and allowing practitioners to focus on model innovation and defect-pattern analysis.

Methods

The HRP4K dataset is constructed using a human-in-the-loop pipeline that emphasizes reproducibility, procedural consistency, and AI-assisted efficiency. As summarized in Fig. 2, the workflow comprises video capture and frame extraction, automated deduplication and quality filtering, privacy-preserving anonymization, and iterative AI-assisted annotation. Each stage incorporates human validation and multi-tier quality control to ensure high-fidelity, publication-ready annotations. Detailed descriptions of each step are provided to ensure transparent reporting of the dataset’s characteristics and provenance.



Fig. 1 Representative examples of high-resolution pothole imagery from the dataset. Green boxes show annotated bounding boxes; blue dashed rectangles mark the zoomed-in areas.

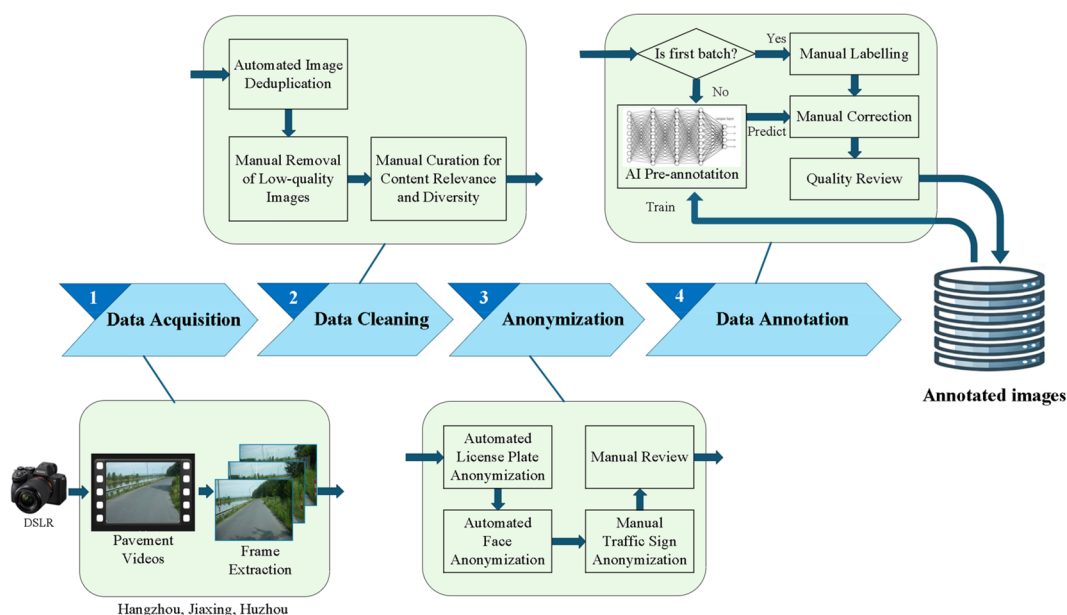


Fig. 2 Dataset generation workflow from data acquisition to final annotation and verification.

Data acquisition. High-resolution road-surface imagery is captured using Sony Alpha 7 IV and Alpha 9 III full-frame mirrorless cameras equipped with a Sony FE 35 mm f/1.8 prime lens. The Alpha 7 IV features a 33-megapixel back-illuminated Exmor R CMOS sensor with strong dynamic range and low-light performance, whereas the Alpha 9 III employs a 24.6-megapixel global-shutter sensor that suppresses rolling-shutter artefacts and is well suited to dynamic on-road capture. To ensure consistency across the entire collection campaign, camera parameters are kept fixed. Video is recorded in XAVC S-I 4K (3840 × 2160 pixels) using All-Intra compression. This compression scheme encodes each frame independently, ensuring that extracted frames retain maximum per-frame quality without inter-frame dependencies. Manual focus is used to maintain a constant focal plane on the road surface and to eliminate autofocus hunting. The aperture is fixed between f/6.3 and f/7.1 to balance depth of field with light throughput. Shutter speed, ISO, and white balance are the only parameters adjusted automatically *in situ*, ensuring proper exposure under varying ambient lighting conditions.



Fig. 3 Pavement data acquisition setup. (a) Data collection vehicle with mounted equipment. (b) Close-up of the camera system.



Fig. 4 Representative examples from the HRP4K dataset. (a) Asphalt pavement. (b) Concrete pavement.

As illustrated in Fig. 3, cameras are mounted on four vehicle types, including one sedan and three commercial vans. Vibration is minimized using only the camera's built-in image stabilisation. This configuration provides a consistent, forward-facing, oblique perspective of the pavement surface, representative of real-world road-monitoring conditions. Data are collected using the four vehicle platforms described above. The cameras are mounted at a height of 1.55–1.70 m above the ground. The central pavement region typically appears at a forward distance of approximately 3.0–4.5 m from the camera, yielding an estimated downward pitch angle of approximately 19–30° relative to the horizontal. Minor deviations arise from road grade and vehicle dynamics.

The survey routes cover approximately 1,100 kilometers of roadways in and around Hangzhou, Jiaxing, and Huzhou (Zhejiang Province, China), spanning multi-lane urban highways, residential streets, and rural roads. Both asphalt and concrete pavements are surveyed, as shown in Fig. 4. Data collection is performed exclusively under daylight and predominantly dry conditions to ensure high image quality. Finally, a custom OpenCV-based program¹⁷ is used to systematically extract frames from the recorded 4 K videos at 3 frames per second (fps), producing an initial pool of several million candidate frames for subsequent processing.

Data cleaning. To ensure dataset quality and relevance, we implement a multi-step data cleaning procedure, as illustrated in Fig. 2 (Step 2). This process begins with automated image deduplication to identify and remove redundant or near-duplicate frames. Here, we employ VisiPics software¹⁸, with the filter option set to Basic and all other settings left at the default values. For each set of duplicate or near-duplicate images identified by VisiPics, we retain the earliest frame and remove the subsequent redundant frames.

Subsequently, the dataset undergoes two stages of manual review based on predefined criteria. In the first stage, we discard low-quality images, including those affected by (i) severe motion blur that prevents clear delineation of pothole boundaries, (ii) extreme overexposure or underexposure that obscures pavement details, (iii) strong glare or reflections that mask the road surface, and (iv) insufficient or obstructed pavement views, for example when the pavement occupies only a small portion of the frame or is largely occluded. In the second stage, we filter the remaining images to ensure relevance and sufficient visual diversity. For positive samples, each image is required to contain at least one visible pothole, while ensuring a diverse representation of pothole sizes, shapes, and severities. Particular attention is also given to achieving a balanced distribution across different pavement textures and surrounding environments. After this process, 4,003 positive high-quality images are retained.

For negative samples, we follow the same quality-control procedure and then curate pothole-absent frames from the raw videos to enable false-positive evaluation under negative-dominant deployment conditions. Specifically, we uniformly sample candidate frames by extracting every 100th frame. To ensure clean negative labels, we remove all positive frames and exclude a temporal buffer of ± 10 frames around each positive frame. The remaining candidates are further deduplicated via VisiPics and manually screened to confirm the absence of potholes, with all ambiguous cases removed.

Anonymization. To protect personal privacy and geographic information, all imagery undergoes a comprehensive anonymization workflow before public release. As shown in Fig. 2 (Step 3), the process begins with automated detection and masking of personally identifiable content. Specifically, facial regions are detected and masked using the YOLOv11 model¹⁹ trained on the WIDERFace dataset²⁰. Concurrently, vehicle license plates are obscured using the YOLOv11 model trained on a license plate recognition dataset²¹. Following this automated pass, all visible traffic signs are manually selected using the LabelMe tool²² and subsequently masked using a custom-developed pixelation filter. This step is performed to remove potentially identifying information that could reveal geographic location.

Finally, two trained reviewers perform a comprehensive manual inspection to verify anonymization effectiveness, ensuring that all privacy-sensitive content is completely obscured. This hybrid automated–manual workflow is essential for maintaining privacy compliance and minimizing human workload.

Data annotation. A semi-automated, iterative annotation workflow, as shown in Fig. 2 (Step 4), is implemented to balance efficiency with the need for high-fidelity labels. The annotation schema comprises a single object class (pothole), defined by a bounding box. The unlabelled, anonymized positive images are partitioned into four sequential batches of approximately 1,000 images each. The process begins with fully manual annotation of the first batch using the LabelMe tool. To ensure consistency and objectivity, we adopt the following annotation guidelines: (i) boxes must strictly enclose the visible pothole boundary while avoiding inclusion of unrelated background; (ii) only visible portions of occluded potholes are labelled, and water-filled or shadowed potholes are included only if the depression boundary is visually distinct; and (iii) adjacent or merged potholes with visible separating pavement are annotated as distinct instances, while continuous clusters are grouped. We apply these rules during cross-annotator review and refinement. As a quality-control step, a senior annotator conducts a second-pass review and refines all annotations in the first batch to establish a high-quality reference set.

This curated first batch is used to train a preliminary YOLOv11 model, which then generates pre-annotations for the second batch. These machine-generated labels are subsequently reviewed and corrected by human annotators, who refine bounding boxes, remove false positives, and add any missed pothole instances. Each corrected image then undergoes the same expert verification as in the first batch. This iterative ‘train-predict-correct-verify’ cycle is repeated for the remaining batches, with the training set incrementally expanded at each stage. For instance, the model used to pre-annotate the third batch is trained on the combined verified labels from the first and second batches. This iterative human-in-the-loop process progressively improves the model’s pre-annotation accuracy, reducing the manual labelling workload by approximately 50% compared with a fully manual approach, while maintaining rigorous annotation quality.

Data Records

The HRP4K dataset is publicly available on the Zenodo repository²³ and is organized to support common object detection frameworks.

Dataset splits. The dataset is divided into training, validation, and test subsets with a 70:15:15 split. To prevent information leakage between data partitions, we perform video-level splitting so that all frames from the same source video are assigned to a single subset. For positive samples, the 4,003 images are derived from 170 distinct video clips. The source video identifier, extracted from the image filename prefix, serves as the unique group ID. Because the number of extracted frames varies across videos, we approximate the target ratio via a deterministic greedy procedure. Video groups are sorted by descending size and sequentially allocated to the subset that minimizes the absolute deviation from the target image count.

Negative samples follow the same video-level grouping to preserve split independence. Within each partition, we then randomly sample 1,400, 300, and 300 pothole-absent images to form the negative train, validation, and test subsets, respectively. Finally, the positive and negative samples are merged to yield a final dataset consisting of 4,203 training, 900 validation, and 900 test images.

Structure. The data are structured in a layout that is directly compatible with both YOLO and COCO-based pipelines. The complete directory structure is illustrated in Fig. 5. The root directory, HRP4K, is partitioned

HRP4K

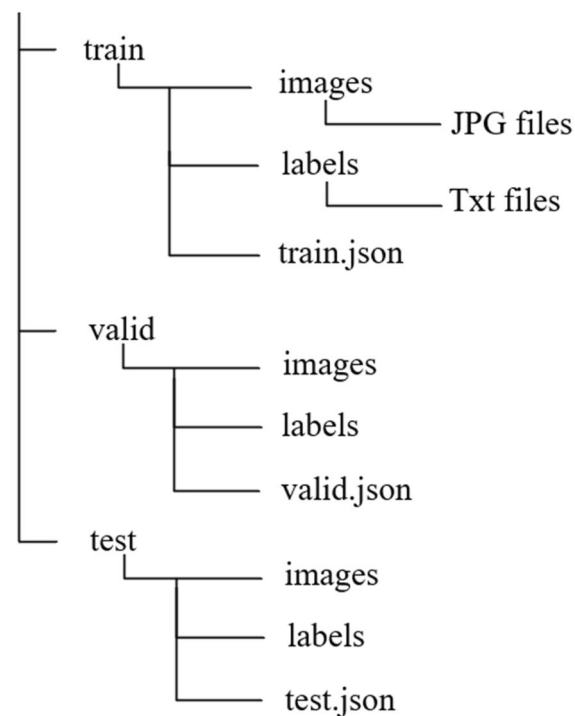


Fig. 5 Directory structure of the HRP4K dataset.

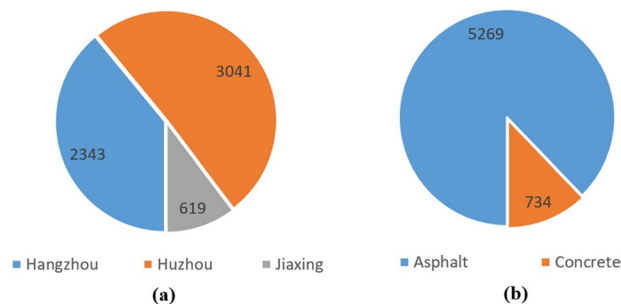


Fig. 6 Geographic and pavement type distributions. **(a)** Geographic distribution of image counts across cities. **(b)** Pavement type distribution. Labels on wedges indicate counts.

into three primary subdirectories: train, valid, and test, corresponding to the training, validation, and test sets, respectively. Each partition follows an identical subdirectory structure. The images directory contains all input road-surface images in JPEG (.jpg) format, while the labels directory stores the corresponding annotation files in YOLO (.txt) format. Each image file is paired with a text file sharing the same base name. This structure enables seamless integration with standard YOLO training and validation pipelines. For COCO-based workflows, each partition includes a consolidated COCO-format JSON file at its root. Training annotations are provided in train.json, and the validation and test partitions follow the same convention (valid.json and test.json) to support standardized benchmarking.

Data Overview

Dataset attributes and statistics. The HRP4K dataset comprises 6,003 images collected across multiple urban and rural environments in Zhejiang Province, China. As illustrated in Fig. 6(a), the images are acquired from three distinct locations: Hangzhou (2,343 images, 39.0%), Huzhou (3,041 images, 50.7%), and Jiaxing (619 images, 10.3%). Figure 6(b) shows the distribution by pavement type. Asphalt surfaces dominate the dataset, comprising 5,269 images (87.8%), whereas concrete pavements account for 734 images (12.2%). This composition reflects the regional predominance of asphalt roads and ensures the representativeness of real-world conditions.

The HRP4K dataset comprises 4,003 positive and 2,000 negative samples. The positive subset contains a total of 7,217 annotated pothole instances. As shown in Fig. 7, the instance distribution exhibits a pronounced long-tail pattern, indicating that most images contain only a few potholes. Specifically, most images (2,594; 64.8%)

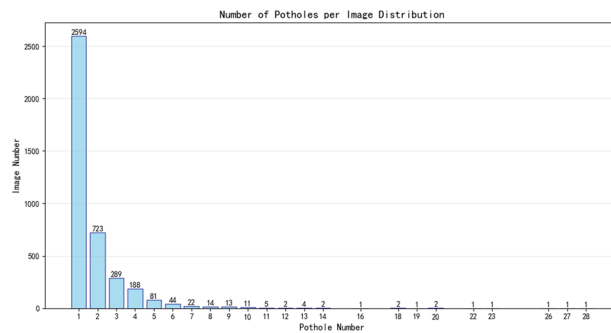


Fig. 7 Distribution of pothole instances per image.

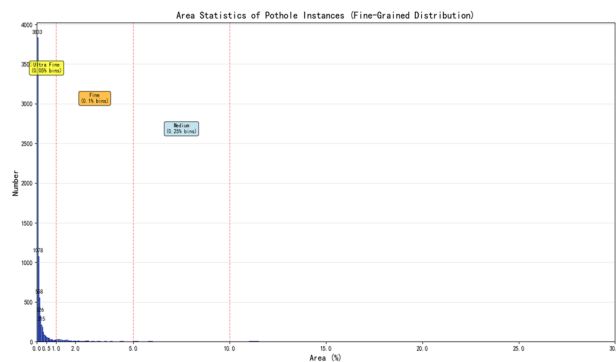


Fig. 8 Distribution of pothole bounding-box area as a percentage of total image area (visual scale).

contain a single pothole, providing clear, isolated examples. A substantial portion (1,200; 30.0%) depicts more complex scenes with two to four potholes, while a small minority (209; 5.2%) contains five or more. Only one image contains the maximum observed count of 28 potholes, highlighting the rarity of dense clusters. This long-tailed distribution reflects the diversity and realism of actual pavement conditions, ensuring that HRP4K can support the development and evaluation of detection algorithms under both sparse and dense object configurations.

We characterize pothole instance scale using the relative bounding-box area, computed as

$$r = \frac{A_{bbox}}{A_{img}} \times 100\%,$$

where $A_{bbox} = w \times h$ is the pothole bounding-box area (in pixels) and $A_{img} = W \times H$ is the image area (in pixels), respectively. As shown in Fig. 8, the distribution of r is highly right-skewed, indicating the dominance of small potholes. To improve the interpretability of this heavy-tailed distribution, we group instances into four visual-scale bins: ultra-fine ($r < 0.05\%$), fine ($0.05\% \leq r < 0.1\%$), medium ($0.1\% \leq r < 0.25\%$), and large ($r \geq 0.25\%$). The cut points (0.05%, 0.1%, 0.25%) are chosen after inspecting the empirical distribution of r and ensuring each bin remains sufficiently populated for stable reporting. Most instances (3,833) fall within the ultra-fine category ($<0.05\%$ of image area), followed by 1,078 in the fine range (0.05–0.1%) and 1099 in the medium range (0.1–0.25%). The remaining 1,207 instances exceed 0.25% of the image area. Notably, r represents the visual scale in the image plane due to the perspective view of the HRP4K dataset. In the absence of camera calibration or depth information, this metric does not directly correspond to actual physical dimensions. Potholes covering more than 1% of the image area are extremely rare, reflecting the natural predominance of small- and medium-scale potholes in urban and suburban roads. This realistic long-tailed distribution of small objects poses a significant challenge for detection models, emphasizing the importance of small-object performance in real-world pavement inspection.

We further characterize the absolute pixel dimensions of the 7,217 pothole instances to benchmark the resolution requirements for detectors. The distribution of bounding box dimensions reveals a significant prevalence of horizontally elongated and ultra-small targets. Specifically, the median bounding box width is 100 pixels and the median height is 35 pixels. Given the 3840×2160 image resolution, these median dimensions occupy only roughly 2.6% of the frame width and 1.6% of the frame height, respectively. Furthermore, the dataset exhibits extreme scale variation, with instances ranging from minute targets approximately 10×4 pixels to massive defects exceeding 3,000 pixels in width. These absolute pixel statistics highlight the stringent requirements for small object detection capabilities in the HRP4K benchmark.

Family	Method	Params	Precision	Recall	FPPI	F1	mAP@0.5	mAP@0.5:0.95
YOLO	YOLOv5	35.7 M	0.651	0.430	0.050	0.518	0.555	0.343
	YOLOv8	25.9 M	0.648	0.466	0.113	0.542	0.580	0.390
	YOLOv11	20.1 M	0.742	0.422	0.030	0.538	0.596	0.407
RT-DETR	RT-DETRv1	36.4 M	0.641	0.560	0.140	0.598	0.606	0.370
	RT-DETRv2	36 M	0.621	0.570	0.113	0.594	0.605	0.373
	D-FINE	19.2 M	0.694	0.520	0.130	0.595	0.611	0.383

Table 2. Performance comparison of object detection methods on HRP4K.

Technical Validation

The HRP4K dataset poses significant challenges, as images are captured in a perspective-view using vehicle-mounted mirrorless cameras at long stand-off distances. Consequently, the images inevitably contain substantial background clutter such as buildings, vehicles, trees, and pedestrians. We conducted multiple validation studies to provide objective evidence of dataset quality and reliability.

Inter-annotator consistency. To assess consistency, three annotators independently labelled a random 200-image subset (~5%), each blinded to the others' labels. Agreement was measured using the Intersection-over-Union (IoU) metric²⁴ with one-to-one matching via the Hungarian algorithm at $\text{IoU} \geq 0.5$. The mean IoU across all pairs of annotators was 0.81 (95% CI), computed via non-parametric bootstrap resampling with 1,000 iterations, which indicates clear and consistently applied guidelines.

Performance of object detection algorithms on HRP4K. We benchmarked six state-of-the-art object detection models on the HRP4K dataset, including YOLOv5²⁵, YOLOv8²⁶, YOLOv11²⁷, RT-DETRv1²⁸, RT-DETRv2²⁹, and D-FINE³⁰. For this comparison, the medium-sized variants were selected. As shown in Table 2, these models represent two distinct architectural philosophies. The former, the YOLO series, is a one-stage dense detector with a CNN backbone and FPN/PAN neck. It treats object detection as a regression and classification problem, applying it to a dense grid of guesses and relying heavily on non-maximum suppression (NMS). In contrast, the RT-DETR family is transformer-based, set-prediction detectors that use sparse queries and one-to-one matching. We report standard metrics, including precision, recall, F1-score, mAP@0.5 (PASCAL VOC; $\text{IoU} = 0.5$), and the COCO primary metric mAP@0.5:0.95 (AP averaged over $\text{IoU} = 0.50:0.95$ in 0.05 increments). To assess the false-alarm rate in negative-dominant deployment scenarios, we evaluated the models specifically on the 300 negative images in the test subset using the false positives per image (FPPI) metric. FPPI is a standard metric in object detection that quantifies the average number of false alarms triggered per input image. These metrics provide a multifaceted evaluation of the model's accuracy for the single pothole class.

All experiments were run on a workstation equipped with four NVIDIA L40S GPUs (48 GB each) and two Intel Xeon 4416 + CPUs under Ubuntu 22.04. We trained all models for 150 epochs with the official default hyperparameters and COCO-pretrained weights, to ensure reproducibility and a fair baseline. The benchmark results are summarized in Table 2. All six models successfully converged and achieved robust performance, confirming that the HRP4K dataset provides a high-quality and consistent signal for training pothole detection models. Among them, D-FINE achieved the highest mAP@0.5 of 0.611, demonstrating strong capabilities in identifying pothole instances despite having the fewest parameters (19.2 M). D-FINE attained improved localization accuracy via fine-grained distribution refinement (FDR), which iteratively models bounding box edges as probability distributions to more precisely capture geometrically irregular and ambiguous objects like potholes. Meanwhile, the transformer-based RT-DETRv1 exhibited the best F1-score of 0.598 and a competitive mAP@0.5 of 0.606. Notably, YOLOv11 demonstrated superior performance in high-precision metrics. It recorded the highest precision of 0.742, the lowest false-alarm rate with an FPPI of 0.030, and the best mAP@0.5:0.95 of 0.407. This indicated that while D-FINE and RT-DETR are highly effective at general detection (mAP@0.5), YOLOv11 generates the tightest bounding boxes and fewest false positives. The widely used YOLOv5 and YOLOv8 models showed comparatively lower performance, with YOLOv5 recording an F1-score of 0.518 and mAP@0.5 of 0.555.

A key observation is the gap between the mAP@0.5 scores and the more stringent mAP@0.5:0.95 scores (e.g., 0.611 vs. 0.383 for D-FINE). This discrepancy is not a flaw, but rather a direct technical validation of the dataset's challenging nature, as detailed in the Dataset Statistics section. HRP4K is dominated by 'ultra-fine' and small pothole instances; for such objects, small pixel-level errors in box localization cause a substantial drop in IoU, which are heavily penalized by the mAP@0.5:0.95 metric.

Pavement-type robustness. The HRP4K dataset contains both asphalt and concrete pavements, presenting distinct challenges for detection models. To quantify material effect, we stratified the test set by pavement type and evaluated the representative CNN-based one-stage detector (YOLOv11) and transformer set-prediction detector (D-FINE). The results, summarized in Table 3, show concrete is markedly more difficult than asphalt for both architectures. Specifically, on asphalt, D-FINE outperforms in terms of F1 (0.636) and mAP@0.5 (0.655), whereas YOLOv11 maintains superior precision (0.762) and mAP@0.5:0.95 (0.436). On concrete, both models degraded, but D-FINE remained ahead: YOLOv11 attained F1 0.284, mAP@0.5 0.385; D-FINE reached F1 0.436, mAP@0.5 0.425.

We attribute the performance gap to two factors. (i) Material imbalance: asphalt images outnumber concrete by approximately $7 \times$ (Fig. 6(b)), which can bias training toward asphalt appearance. (ii) Distinct visual characteristics: concrete pavements show repeating textures and linear joint lines that resemble pothole edges. This

Pavement types	Method	Precision	Recall	FPPI	F1	mAP@0.5	mAP@0.5:0.95
Asphalt	YOLOv11	0.762	0.485	0.033	0.593	0.640	0.436
	D-FINE	0.704	0.580	0.116	0.636	0.655	0.413
Concrete	YOLOv11	0.590	0.187	0.033	0.284	0.385	0.280
	D-FINE	0.494	0.390	0.304	0.436	0.425	0.260

Table 3. Performance comparison of object detection methods on the two pavement types.

complexity significantly increases false positives and localization errors. Conversely, asphalt typically presents more stochastic, irregular crack patterns and patch repairs. This experiment indicates that current detectors still struggle with cross-material generalisation even for the same pothole class, and positions HRP4K as a robust and challenging baseline for evaluating robustness across pavement types.

Influence of the anonymization process. To evaluate the impact of our anonymization techniques on model performance, we conducted a comparative analysis. This process involved two distinct validation stages. First, we established a performance baseline by training the model on the original, non-anonymized dataset and validating its accuracy. Subsequently, we evaluated the same model using the fully anonymized dataset, ensuring all testing conditions remained consistent. The comparative analysis revealed no significant differences in validation results between the anonymized and non-anonymized datasets. This finding indicates that inpainting-based anonymization does not degrade pothole-detection performance and preserves the integrity of the data-processing pipeline.

Usage Notes

The HRP4K dataset is intended to advance research on pothole detection from on-road, perspective-view imagery. This context is critical, as targets in such imagery are typically small, distant, and embedded in visually complex scene. These characteristics present a challenge that prior resources, often limited by scale or coarse resolution, do not adequately address. The dataset provides high-resolution images with single pothole class annotations; other surface distresses (e.g., cracks, rutting) are intentionally not labelled to keep the benchmark unambiguous. Annotations are provided in two standardized formats in both YOLO and COCO formats, as detailed in the Data Records section. This ensures immediate compatibility with common deep-learning pipelines and facilitates direct, reproducible benchmarking. We encourage extending HRP4K with task-specific annotations to support other supervised learning tasks.

The dataset is publicly available under the CC BY 4.0 license.

Limitations:

- Geographic and environmental bias. The dataset is collected exclusively in three cities within Zhejiang Province, China. Despite diverse road types (urban, rural) and pavement materials (asphalt, concrete), it covers a limited climatic and temporal envelope. Models trained solely on HRP4K may require fine-tuning or domain adaptation to generalize to other regions or to adverse weather conditions.
- Perspective bias. All imagery is captured from vehicle-mounted, forward-facing cameras at oblique angles; aerial or nadir-view models may not transfer without adaptation. Pothole scales represent projected visual areas in the image plane, which do not directly correspond to their actual physical dimensions. Methods developed for aerial, nadir-view, or alternative viewpoints may not transfer directly without adaptation.

Data availability

The HRP4K dataset is publicly available in the Zenodo repository²³ at <https://doi.org/10.5281/zenodo.17522874>.

Code availability

Manual bounding-box annotation of potholes is performed with LabelMe (<https://github.com/wkentaro/labelme>). For anonymization and automatic pre-annotation, we use the YOLOv11 model implemented in the Ultralytics library (<https://github.com/ultralytics/ultralytics>). For the technical validation benchmark, models including YOLOv5, YOLOv8, and YOLOv11 are trained and evaluated using the Ultralytics framework. The RT-DETRv1 and RT-DETRv2 are implemented with the RT-DETR framework (<https://github.com/lyuwenyu/RT-DETR>), while the D-FINE model is based on its official implementation (<https://github.com/Peterande/D-FINE>). All custom code developed for this study is publicly available at <https://github.com/hanshenChen/HRP4K>. The repository includes scripts for video frame extraction, privacy anonymization, model-assisted pre-annotation, and annotation format conversion.

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Author contributions

Hanshen Chen conceived the study, secured funding, and designed the methodology. Hanshen Chen and Zhoulin Tu collected the data and implemented the experiments. Hanshen Chen and Yu Zhao performed the analyses and interpreted the results. Jianfeng Ye provided strategic guidance and contributed to discussions. Hanshen Chen drafted the manuscript. All authors reviewed and approved the final version of the paper.

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Competing interests

The authors declare no competing interests.

Additional information

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